

Git and GitHub – Version Control

Description

1. sign up to the GITHUB and create a project repository.
2. install and configure GIT version control system in your local machine.
3. Verify the configurations by cloning the remote repository
4. try to add local files to the remote repository as vice versa.

Note: Learn more about important GIT commands.

1. Git Installation and Configuration

Git for Windows was installed, which provided **Git Bash** for executing Git commands.

The Git installation was verified using:

```
git --version
```

The Git username was configured using:

```
git config --global user.name "Mohammed Sanjid"
```

The GitHub email address was configured using:

```
git config --global user.email "mohammedsanjid77@gmail.com"
```

The configuration was verified using:

```
git config --list
```

Screenshot 1 – Git Installation and Configuration

```
mhdsa@HP-DESKTOP MINGW64 ~
$ git --version
git version 2.55.0.windows.3

mhdsa@HP-DESKTOP MINGW64 ~
$ git config --global user.name "Mohammed Sanjid"

mhdsa@HP-DESKTOP MINGW64 ~
$ git config --global user.email "mohammedsanjid77@gmail.com"

mhdsa@HP-DESKTOP MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
windows.appendatomically=false
user.name=Mohammed Sanjid
user.email=mohammedsanjid77@gmail.com

mhdsa@HP-DESKTOP MINGW64 ~
$ git config --global user.name
Mohammed Sanjid

mhdsa@HP-DESKTOP MINGW64 ~
$ git config --global user.email
mohammedsanjid77@gmail.com
```

2. GitHub Repository Creation

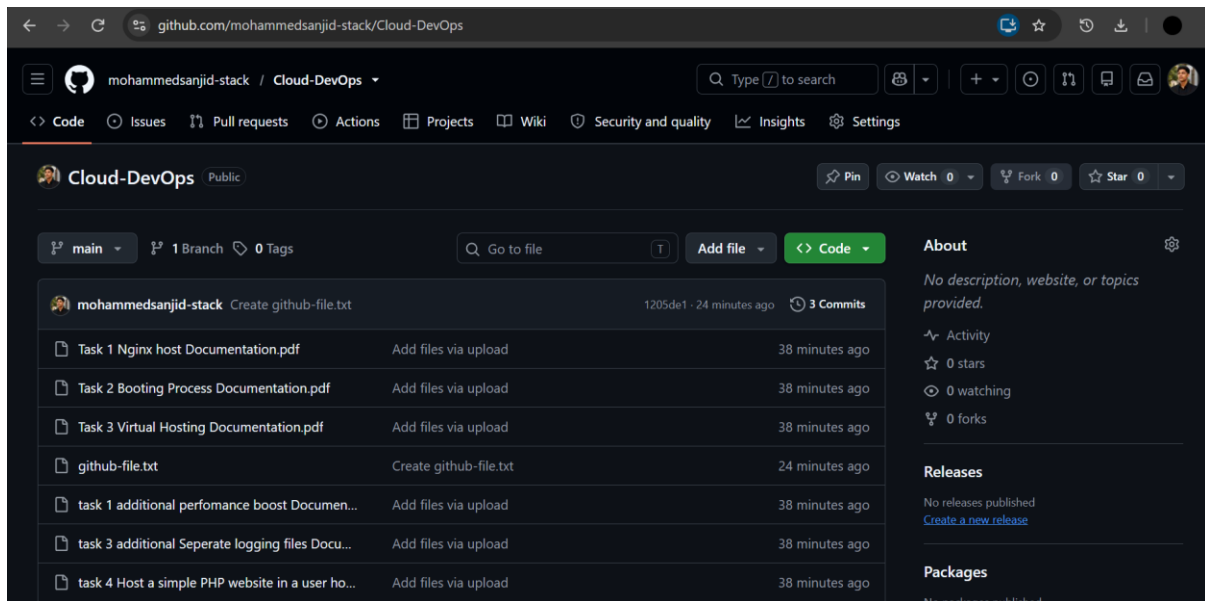
A GitHub repository named **Cloud-DevOps** was created.

Repository URL:

<https://github.com/mohammedsanjid-stack/Cloud-DevOps.git>

The repository was used as the remote repository for the practical.

Screenshot 2 – GitHub Repository



3. Cloning the Remote Repository

The GitHub repository was cloned to the local Windows machine using:

```
git clone https://github.com/mohammedsanjid-stack/Cloud-DevOps.git
```

The cloned repository was accessed using:

```
cd Cloud-DevOps
```

The repository status was checked using:

```
git status
```

The output confirmed that the local repository was on the main branch and was synchronized with the remote repository.

Screenshot 3 – Repository Cloning and Verification

```
mhdas@HP-DESKTOP MINGW64 ~/Desktop
$ git clone https://github.com/mohammedsanjid-stack/Cloud-DevOps.git
Cloning into 'Cloud-DevOps'...
remote: Enumerating objects: 25, done.
remote: Counting objects: 100% (25/25), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 25 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (25/25), 11.48 MiB | 9.18 MiB/s, done.
```

4. Verifying the Remote Repository

The connection between the local repository and GitHub was verified using:

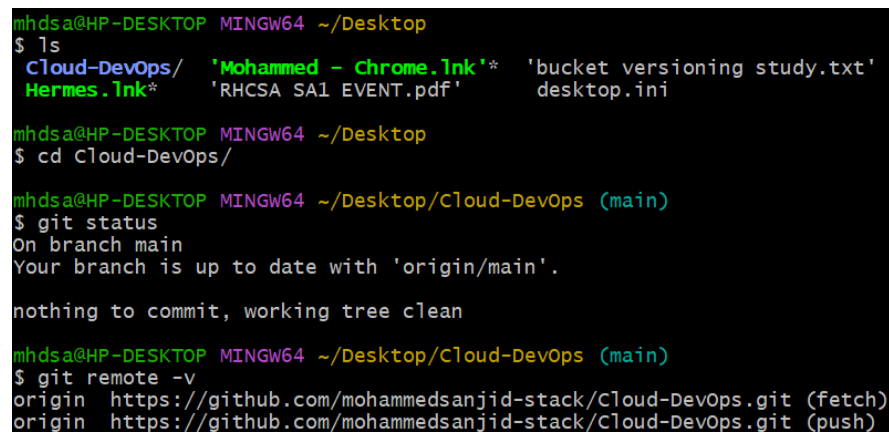
```
git remote -v
```

The output showed:

```
origin  https://github.com/mohammedsanjid-stack/Cloud-DevOps.git (fetch)
origin  https://github.com/mohammedsanjid-stack/Cloud-DevOps.git (push)
```

This confirmed that the local Git repository was connected to the GitHub repository.

Screenshot 4 – Remote Repository Verification



```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop
$ ls
Cloud-DevOps/ 'Mohammed - Chrome.Ink'* 'bucket versioning study.txt'
Hermes.Ink* 'RHCSA SA1 EVENT.pdf' desktop.ini

mhdsa@HP-DESKTOP MINGW64 ~/Desktop
$ cd Cloud-DevOps/

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git remote -v
origin  https://github.com/mohammedsanjid-stack/Cloud-DevOps.git (fetch)
origin  https://github.com/mohammedsanjid-stack/Cloud-DevOps.git (push)
```

5. Adding a Local File to Git

A local file named **test.txt** was created inside the cloned repository.

The file was added to the Git staging area using:

```
git add test.txt
```

The staging status was verified using:

```
git status
```

The output showed **test.txt** under **Changes to be committed**.

Screenshot 5 – Adding the Local File

```
mhdasa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ echo "Git Practice Completed" > test.txt

mhdasa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ ls
'Task 1 Nginx host Documentation.pdf'
'Task 2 Booting Process Documentation.pdf'
'Task 3 Virtual Hosting Documentation.pdf'
'task 1 additional performance boost Documentation.pdf'
'task 3 additional Seperate logging files Documentation.pdf'
'task 4 Host a simple PHP website in a user home directory documentation.pdf'
'task-10 ebs and efs.pdf'
'task-11 wordpress hosting and rds.pdf'
'task-12 Implement NAT Gateway Setup.pdf'
'task-13 backup script with S3 bucket.pdf'
'task-14 webhosting using S3 bucket.pdf'
'task-15 server monitoring tool.pdf'
'task-16 Troubleshooting and security implementation in Linux.pdf'
'task-17 Disk space utilization notification using SNS.pdf'
'task-18 AWS SES Email Sending Using PHP.pdf'
'task-19 server tuning.pdf'
'task-20 Web Hosting and DNS.pdf'
'task-21 load balancer and auto scaling.pdf'
'task-5 Dynamic webpage hosting.pdf'
'task-6 Website Hosting Errors.pdf'
'task-7 file transfer using sftp & vsftp.pdf'
'task-8 iam user lambda,cloudwatch.docx'
'task-9 iaam user restrictions.docx'
test.txt

mhdasa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   test.txt
```

6. Committing the Local File

The staged file was committed to the local Git repository using:

```
git commit -m "Add test file"
```

The commit was successfully created.

The commit recorded the changes locally before they were uploaded to GitHub.

Screenshot 6 – Git Commit

```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git commit -m "Add text file"
[main 7839e91] Add text file
1 file changed, 1 insertion(+)
create mode 100644 test.txt
```

7. Pushing the File to GitHub

The committed changes were uploaded to the GitHub repository using:

```
git push origin main
```

After the push operation, the `test.txt` file was visible in the GitHub repository.

This verified the **local-to-remote** Git workflow.

Screenshot 7 – Push to GitHub

```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 302.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mohammedsanjid-stack/Cloud-DevOps.git
bd0172e..7839e91 main -> main
```

8. Pulling Changes from GitHub

To verify the reverse operation, a file named **github-file.txt** was created directly in the GitHub repository.

The latest changes were downloaded to the local repository using:

```
git pull origin main
```

The local files were checked using:

```
ls
```

The downloaded `github-file.txt` file was visible in the local repository.

This verified the **remote-to-local** Git workflow.

Screenshot 8 – Pulling Changes from GitHub

```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ git pull origin main
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 957 bytes | 50.00 KiB/s, done.
From https://github.com/mohammedsanjid-stack/Cloud-DevOps
 * branch            main       -> FETCH_HEAD
    7839e91..1205de1  main       -> origin/main
Updating 7839e91..1205de1
Fast-forward
 github-file.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 github-file.txt

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)
$ ls
'Task 1 Nginx host Documentation.pdf'
'Task 2 Booting Process Documentation.pdf'
'Task 3 Virtual Hosting Documentation.pdf'
github-file.txt
'task 1 additional performance boost Documentation.pdf'
'task 3 additional Seperate logging files Documentation.pdf'
'task 4 Host a simple PHP website in a user home directory documentation.pdf'
'task-10 ebs and efs.pdf'
'task-11 wordpress hosting and rds.pdf'
'task-12 Implement NAT Gateway Setup.pdf'
'task-13 backup script with S3 bucket.pdf'
'task-14 webhosting using S3 bucket.pdf'
'task-15 server monitoring tool.pdf'
'task-16 Troubleshooting and security implementation in Linux.pdf'
'task-17 Disk space utilization notification using SNS.pdf'
'task-18 AWS SES Email Sending Using PHP.pdf'
'task-19 server tuning.pdf'
'task-20 Web Hosting and DNS.pdf'
'task-21 load balancer and auto scaling.pdf'
'task-5 Dynamic webpage hosting.pdf'
'task-6 Website Hosting Errors.pdf'
'task-7 file transfer using sftp & vsftp.pdf'
'task-8 iam user lambda,cloudwatch.docx'
'task-9 iaam user restrictions.docx'
test.txt
```

9. Git Repository Collaboration

To verify Git collaboration, the repository of another GitHub user was used. The repository was accessed locally from:

`~/Desktop/Cloud-DevOps/my-repo`

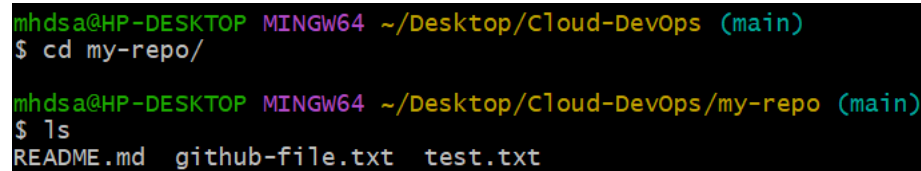
The repository was connected to the friend's GitHub repository:

<https://github.com/shefinroshan7/my-repo>

The repository initially contained:

```
README.md  
github-file.txt  
test.txt
```

Screenshot 9 – Friend's Repository



```
mhd5a@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps (main)  
$ cd my-repo/  
  
mhd5a@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)  
$ ls  
README.md  github-file.txt  test.txt
```

10. Pulling a File Added by the Collaborator

The collaborator added a file named `shefin.txt` to the GitHub repository.

The latest changes were downloaded using:

```
git pull origin main
```

The pull operation completed successfully:

```
Updating 5a3b038..4956362  
Fast-forward  
  shefin.txt | 1 +  
  1 file changed, 1 insertion(+)  
  create mode 100644 shefin.txt
```

The local repository was then checked using:

```
ls
```

The `shefin.txt` file was successfully downloaded and appeared in the local repository.

The local repository then contained:

```
README.md  
github-file.txt  
shefin.txt  
test.txt
```

Screenshot 10 – Pulling Collaborator's File

```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ git pull origin main
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 2 (delta 1), reused 2 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 218 bytes | 13.00 KiB/s, done.
From https://github.com/shefinroshan7/my-repo
* branch          main      -> FETCH_HEAD
   5a3b038..4956362  main      -> origin/main
Updating 5a3b038..4956362
Fast-forward
 shefin.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 shefin.txt

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ ls
README.md  github-file.txt  shefin.txt  test.txt
```

11. Creating and Pushing a File to the Collaborator's Repository

A new file named `sanjid.txt` was created locally using:

```
echo "This is sanjid's file" > sanjid.txt
```

The file was added to the Git staging area using:

```
git add sanjid.txt
```

The file was committed using:

```
git commit -m "Add sanjid file"
```

The commit was successfully created:

```
[main b5c591d] Add sanjid file
1 file changed, 1 insertion(+)
 create mode 100644 sanjid.txt
```

The committed file was pushed to the shared GitHub repository using:

```
git push origin main
```

The push was successful:

```
4956362..b5c591d  main -> main
```

The file `sanjid.txt` was therefore uploaded to the collaborator's GitHub repository.

Screenshot 11 – Pushing the File

```
mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$
git add sanjid.txt
warning: in the working copy of 'sanjid.txt', LF will be replaced by CRLF the next time Git touches it

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ git commit -m "Add sanjid file"
[main b5c591d] Add sanjid file
1 file changed, 1 insertion(+)
create mode 100644 sanjid.txt

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ git bush origin main
git: 'bush' is not a git command. See 'git --help'.

The most similar command is
push

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 293.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/shefinroshan7/my-repo.git
4956362..b5c591d  main -> main

mhdsa@HP-DESKTOP MINGW64 ~/Desktop/Cloud-DevOps/my-repo (main)
$ ls
README.md  github-file.txt  sanjid.txt  shefin.txt  test.txt
```

12. Verifying the Collaboration

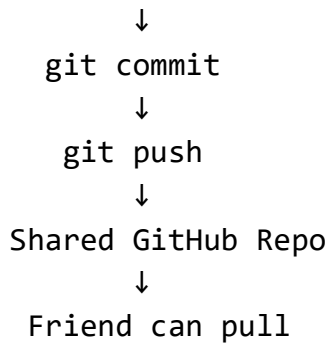
After the push, the local repository contained both files:

```
README.md
github-file.txt
sanjid.txt
shefin.txt
test.txt
```

The collaboration demonstrated two-way file synchronization:

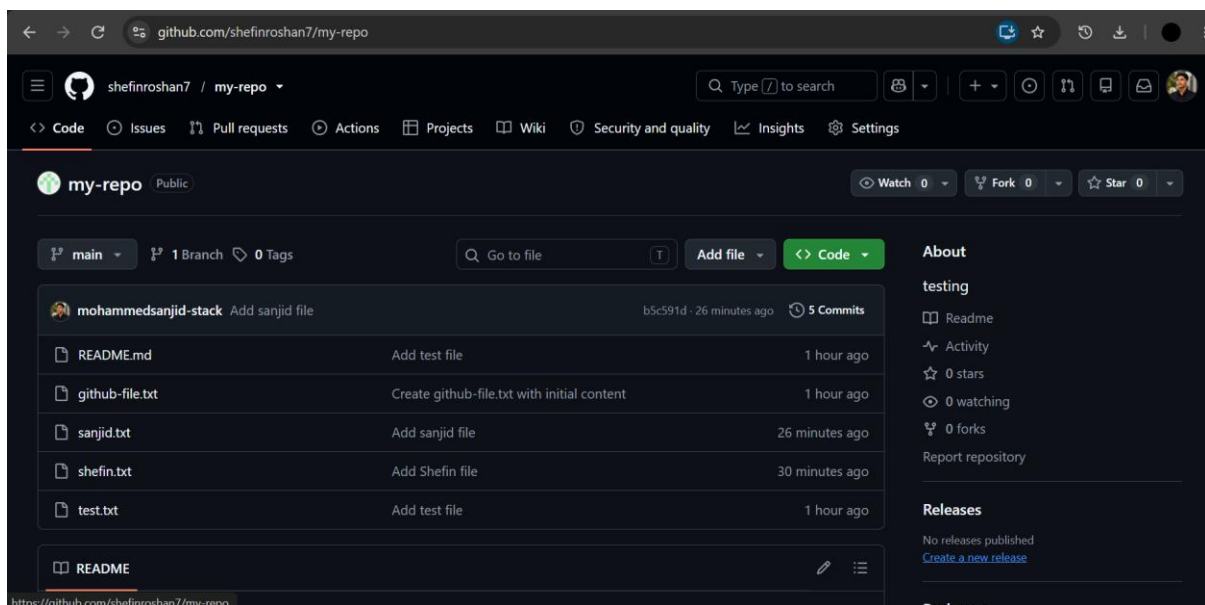
Friend's Local/GitHub Changes

```
↓
git pull
↓
My Local Repo
↓
git add
```



This demonstrated how Git can be used by multiple users to **share, download, modify, and upload files through a common remote repository.**

Screenshot 12 – Final Repository State



Conclusion

Git was successfully installed and configured on the local Windows machine, and the GitHub repository was cloned and connected to the local repository. Local files were added, committed, and pushed to GitHub, while changes made on GitHub were pulled back to the local machine.

Git collaboration was also practiced using a friend's repository. A file added by the collaborator was successfully pulled into the local repository, and a new file was created, committed, and pushed to the shared repository. This demonstrated the basic Git workflow, remote repository synchronization, and collaboration between multiple users using Git and GitHub.

